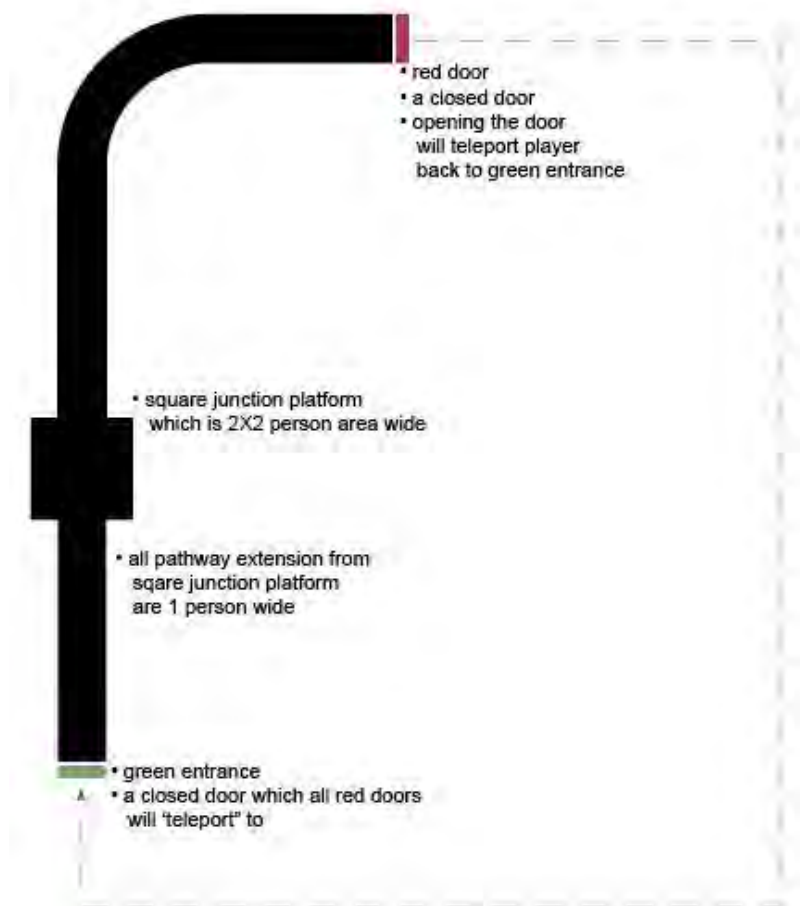


3D Real-Time Diploma in Immersive Media & Game Design Year 2 (2025/26) Semester 2.1	Duration: Week 8 - 13 Due Date: WK13 Sunday 2359hrs.
ASG2: Sci-Fi Corridor – 25%	

1. Overview

You are to build a “sci-fi corridor” environment consisting of a central hero area with a L-shaped junction. On the right is the rough plan view of the scene given by the client.

The detail nature of the pathway relative to each other and to the junction area is up to your creative proposal (e.g. paths need not be of same length, paths could have more challenging elements of stairs up or down, etc.).

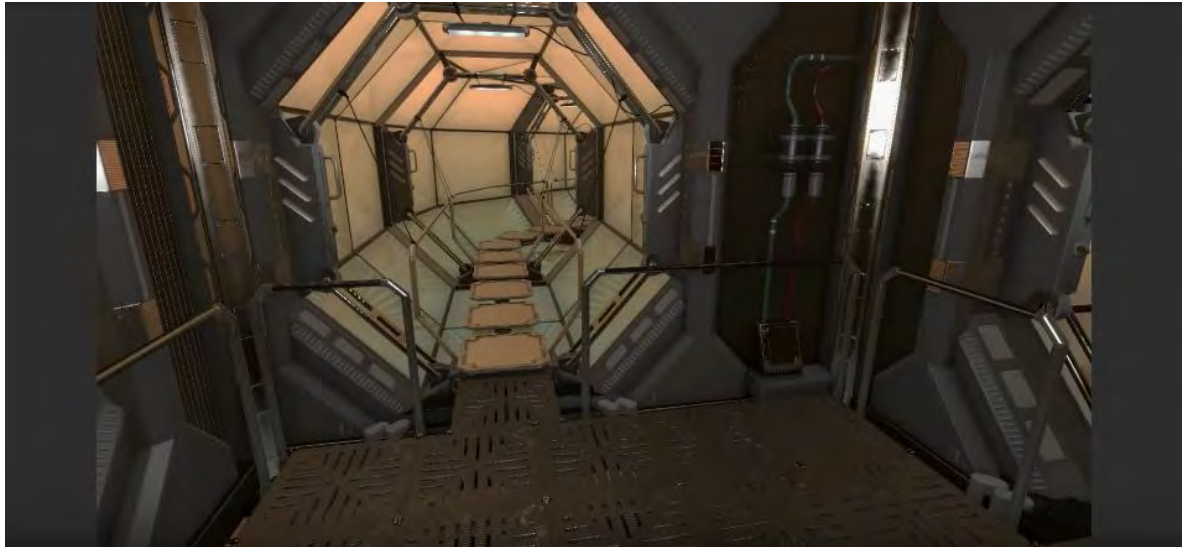


2. Aim

- Apply appropriate workflow for real-time 3D productions.
- Apply appropriate modelling techniques for real-time 3D productions.
- Apply different texturing techniques for real-time 3D productions.
- Apply appropriate lighting for real-time 3D productions.
- Apply post processing to enhance real-time 3D environment.

3. Project Details

3D build a sci-fi interior space with a L-shape corridor consisting of a central hero area.



from Random Seed Game (<https://www.youtube.com/watch?v=bbW44s4MF9k>, from 3:18 mins onwards)

The completed environment should display practical knowledge of:

- modular modelling with topological economy.
- high to low model detail maps baking.
- trim sheet / texture atlas texturing process.
- hero asset unique UV texturing process.
- real-time engine 3D asset assembly, lighting, vfx and post process processing.

The genre of the environment as indicated is sci-fi in nature:



Star Wars 1977 ([Tantive IV Corridor](#))



Alien 1979 ([Nostromo Corridor](#))



Star Wars 1977 ([Detention Block Corridor](#))



Alien 1979 ([Nostromo Corridor](#))

Pitch your idea with background story, visual concepts and/or real-life image references with PureRef for client approval before proceeding to White-boxing and 3D asset building phase.



You are largely working remotely with your client. The onus is on you to initiate communication regularly with your client to get feedbacks at various phases of the workflow to meet client expectations and deadline.

The final deliverable would be a compiled “playable” **first person** camera controller Unity3D build.

4. Deliverables

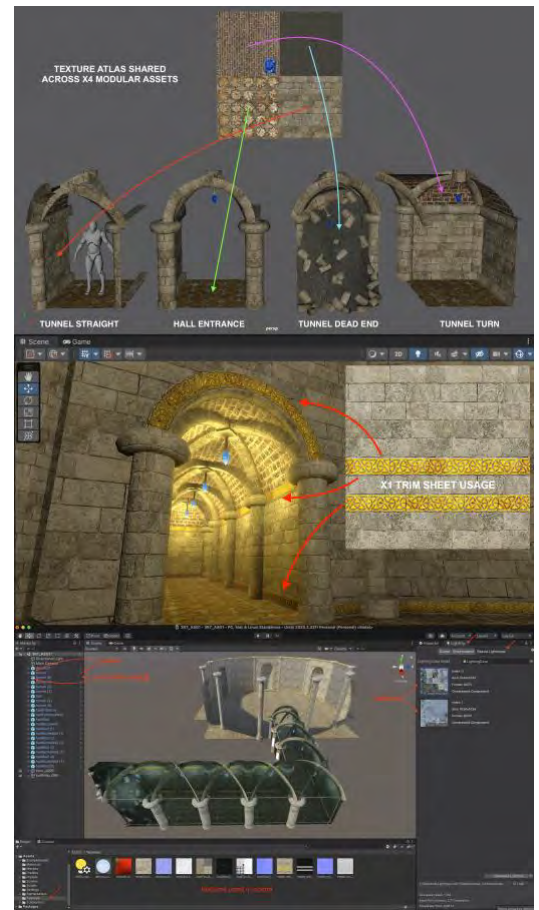
- PureRef visual research mood-board.
- Appropriate VFX (at least 2) elements applied where needed.
- Appropriate lighting done within real-time editor.
- Appropriate post-processing added within real-time editor.
- Appropriate stand-alone “playable” **first person camera controller** build done.
- Unity Project folder of finalized project containing:
 - At least X4 Modular Assets (built from scratch).
 - At least X1 Hero Asset.
 - X4 (at least) modular assets sharing a single Texture Atlas texture set.
 - X1 (at least) sample of Trim Sheet usage.

Deliverables to be submitted to POLITEmail under ASG2 Submission Portal tab. There are X6 separate submission sub-portals with corresponding naming convention as follow:

Sub-portals	Naming Convention
- ASG2 – Texture Atlas/Trim Sheet Still Image Submission - ASG2 – Hero Asset Still-Image N Texture Source Credit Submission - ASG2 – Unity Build Submission - ASG2 – Unity Project Submission - ASG2 – Build Vid Capture Submission - ASG2 – Unity Project Editor Content Vid Capture Submission	<ClassIndex#>_<FullName>_s-image.png <ClassIndex#>_<FullName>_heroAsset.doc <ClassIndex#>_<FullName>_u-build.zip <ClassIndex#>_<FullName>_u-prj.zip <ClassIndex#>_<FullName>_buildVid.mp4 <ClassIndex#>_<FullName>_uPrjVid.mp4

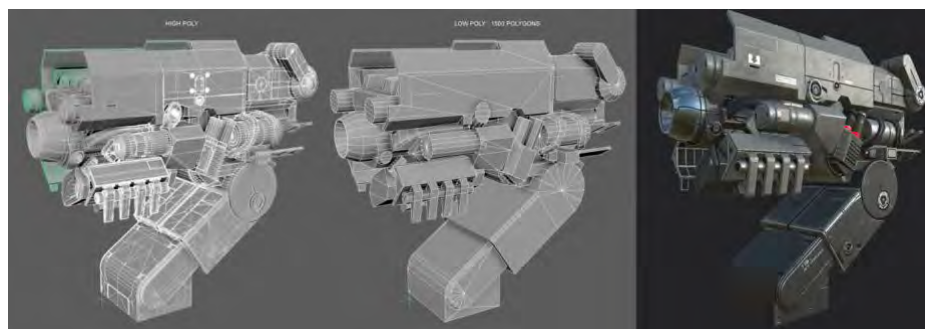
For ASG2 – Still Image Submission, a single image of 1920 pixel in width showing the following are required:

- a single texture atlas that contains textures shared among 4 different modular assets.
- a trim sheet that has at least one trim texture used in texturing modular assets.
- a Unity editor UI screen capture showing:
 - Hierarchy menu containing domineering prefabs (demonstration that modular asset creation is employed).
 - Lighting menu with baked lightmaps.
 - Project menu with Textures employed for the project, making sure there is no unnecessary duplicates of diffuse_cube, ibl_brdf_lut, or textures in general.
- Name the image
<ClassIndex#>_<FullName>_s-image.png



Please see sample image on right.

For the Hero Asset Hero Asset Still-Image N Texture Source Credit Submission, screenshot high, low poly model of your hero asset, and the Substance Painted low poly compiled as a single image like below:



Place the single image in a Word document.

Underneath the image, place credits of texture sources used for ASG3 Project (for modular as well as hero assets texturing) with online links. Name the document

<ClassIndex#>_<FullName>_heroAsset.doc

For ASG2 – Build Vid Capture Submission, get a 1920X1080 resolution MP4 video record of stand-alone “playable” 1st person camera build run clearly showing the following:

- Start video with the clicking of your build .exe file transiting into the “Build with Unity” logo.
- X2 VFX implemented in the scene.
- Appropriate lighting done in the scene.
- Post process (bloom, vignette, fog, etc).
- Functional portal seamless transition from end of walk back to beginning of walk.
- Name video **<ClassIndex>_<FullName>_buildVid.mp4**

For ASG2 – Unity Project Editor Content Vid Capture Submission, get a 1920X1080 resolution MP4 video record of Unity Project clearly showing the following:

- that all non-moving components (including lights) are set to static.
- that all non-moving lights used in scene are set to mixed.
- that Global Volume post processing node is set in hierarchy with proper values set.
- that post process parameter is ticked for game camera.
- that proper prefabing of modular asset is done in layout.
- that there is no unnecessary repeat of textures for the modular assets.
- Name video **<ClassIndex>_<FullName>_UprjVid.mp4**

5. Assessment Criteria

In general, the assessment criteria will encompass the following areas:

- Quality of PureRef Visual Research.
- Whiteboxing.
- X1 Texture atlas shared among at least X4 Modular Assets.
- X1 instance of Trim sheet usage.
- X1 Hero Asset with High Poly to Low Poly mesh Map baking done.
- X2 VFX applied (eg. sparks, steam, dust, etc ...).
- Lighting (with appropriate light map baking).
- Post-Processing.
- Build Success.

6. Late Submission

Late submission will be penalized (10% of the marks for each day late). Submission will not be accepted after 5 days (including weekends and public holidays) from the date of submission.

Reach out to your tutor via email or other communication platform stating lateness reason and arrange for alternate late submission process.

7. Plagiarism and Copyright Issues

Plagiarism means, “copying any part of a source, and then submitting it, claiming that it is your own work.”

Please ensure that all the works submitted by you are not copied from other sources. Any attempt to plagiarize will be dealt with severely, and it may result in your failing the module.

If you have made any references to certain materials, make sure you cite the sources by acknowledging and providing the information necessary to find the source (e.g. Title and author of book, Internet links, etc)

Please refer to the following URL for more details:

<https://www1.np.edu.sg/clte/antiplagiarism/policy.htm>